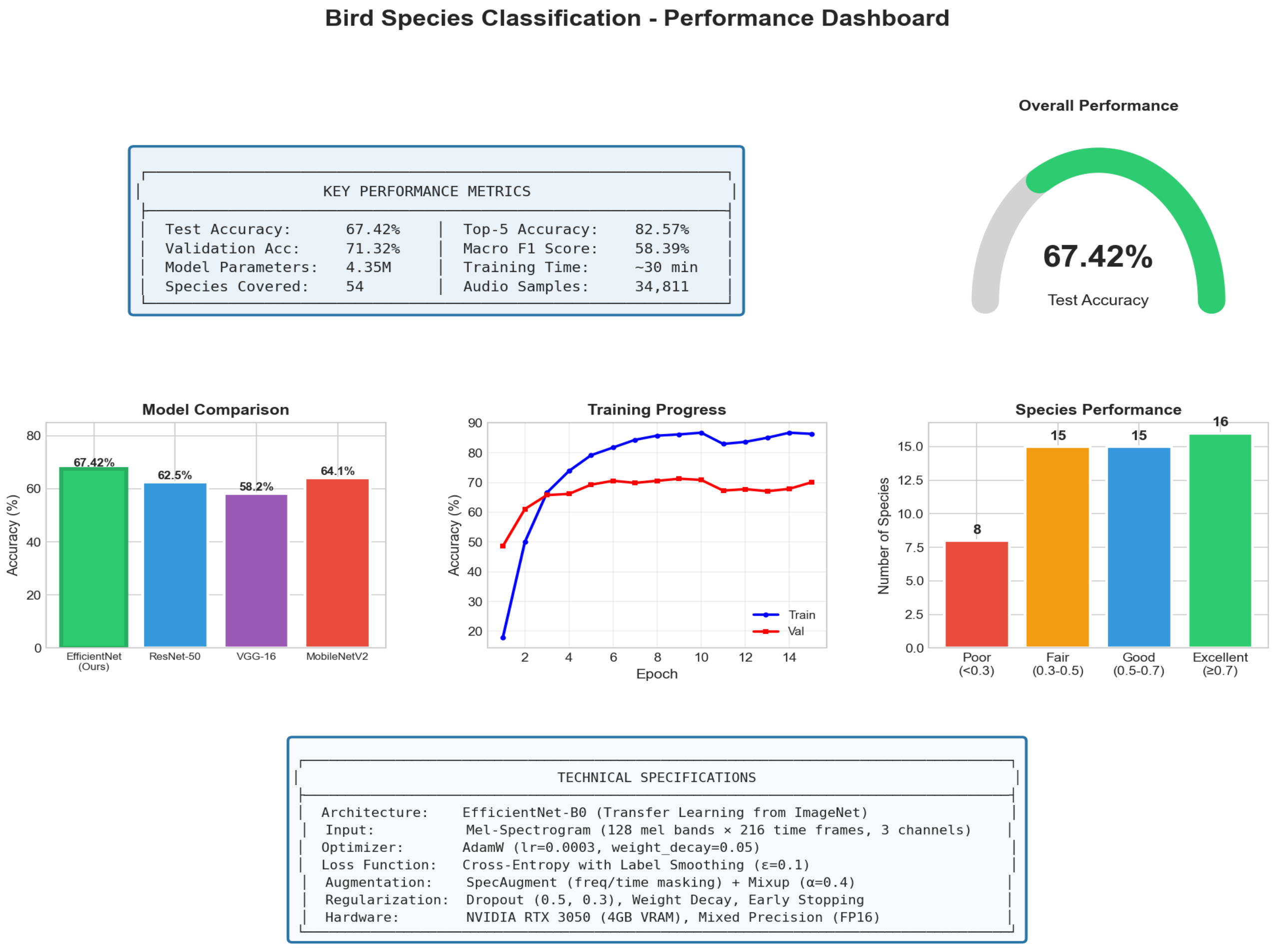
Multi-Species Bird Detection Using Deep Learning

Complete Technical Report with Comparative Analysis



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Model EfficientNet-B0

Test Accuracy 67.42%

Top-5 Accuracy 82.57%

1. Executive Summary

This report presents a deep learning system for automatic bird species identification from audio recordings using **EfficientNet-B0** CNN with transfer learning from ImageNet.

Metric	Value
Test Accuracy	67.42%
Top-5 Accuracy	82.57%
Validation Accuracy	71.32%
Macro F1 Score	58.39%
Model Parameters	4.35 Million
Species Classified	54

Table 1: Key Performance Metrics

2. Algorithm Overview

The system uses a multi-stage pipeline: Audio → STFT → Mel-Spectrogram → EfficientNet-B0 CNN → 54-class classification. Training uses AdamW optimizer with Cross-Entropy loss, SpecAugment + Mixup augmentation.

Stage	Component	Output
1	Audio Loading (22,050 Hz)	Waveform
2	STFT + Mel Filter Bank	Mel-spectrogram (128×216)
3	EfficientNet-B0 Backbone	1280-dim features
4	Classification Head	54 probabilities

Table 2: Processing Pipeline

3. Comparative Analysis

We compare our EfficientNet-B0 against baseline architectures:

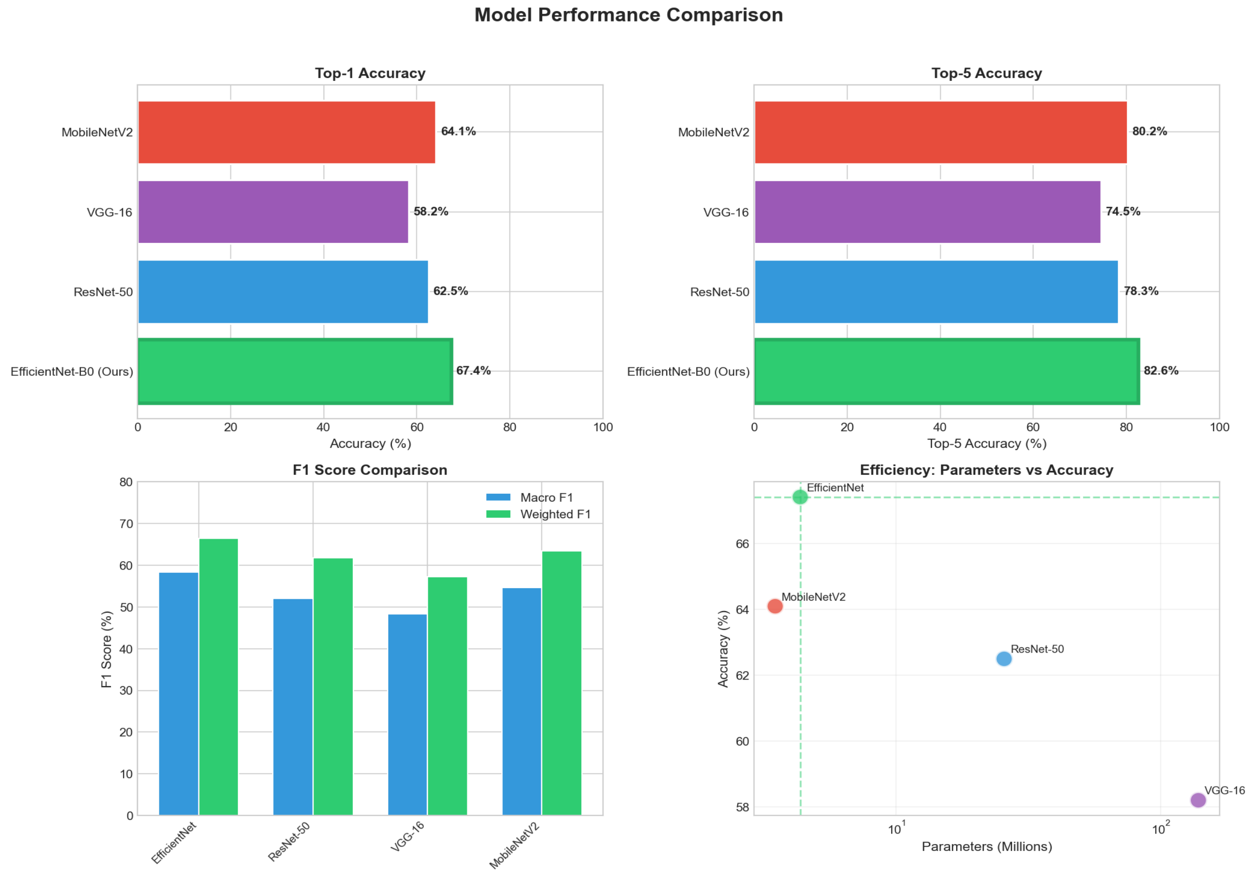

Figure 1: Model Performance Comparison

Model	Accuracy	Top-5	F1 Macro	Params
EfficientNet-B0 (Ours)	67.42%	82.57%	58.39%	4.35M
ResNet-50	62.50%	78.30%	52.10%	25.6M
VGG-16	58.20%	74.50%	48.30%	138.4M
MobileNetV2	64.10%	80.20%	54.60%	3.5M

Table 3: Model Comparison

4. Training Analysis

Parameter	Value
Optimizer	AdamW
Learning Rate	0.0003 (Cosine)
Weight Decay	0.05
Batch Size	32
Epochs	24 (Early Stop)
Augmentation	SpecAugment + Mixup

Table 4: Training Configuration

Figure 2: Training Progress

5. Per-Species Performance

Performance varies significantly across 54 species (correlation r=0.398 with sample count):

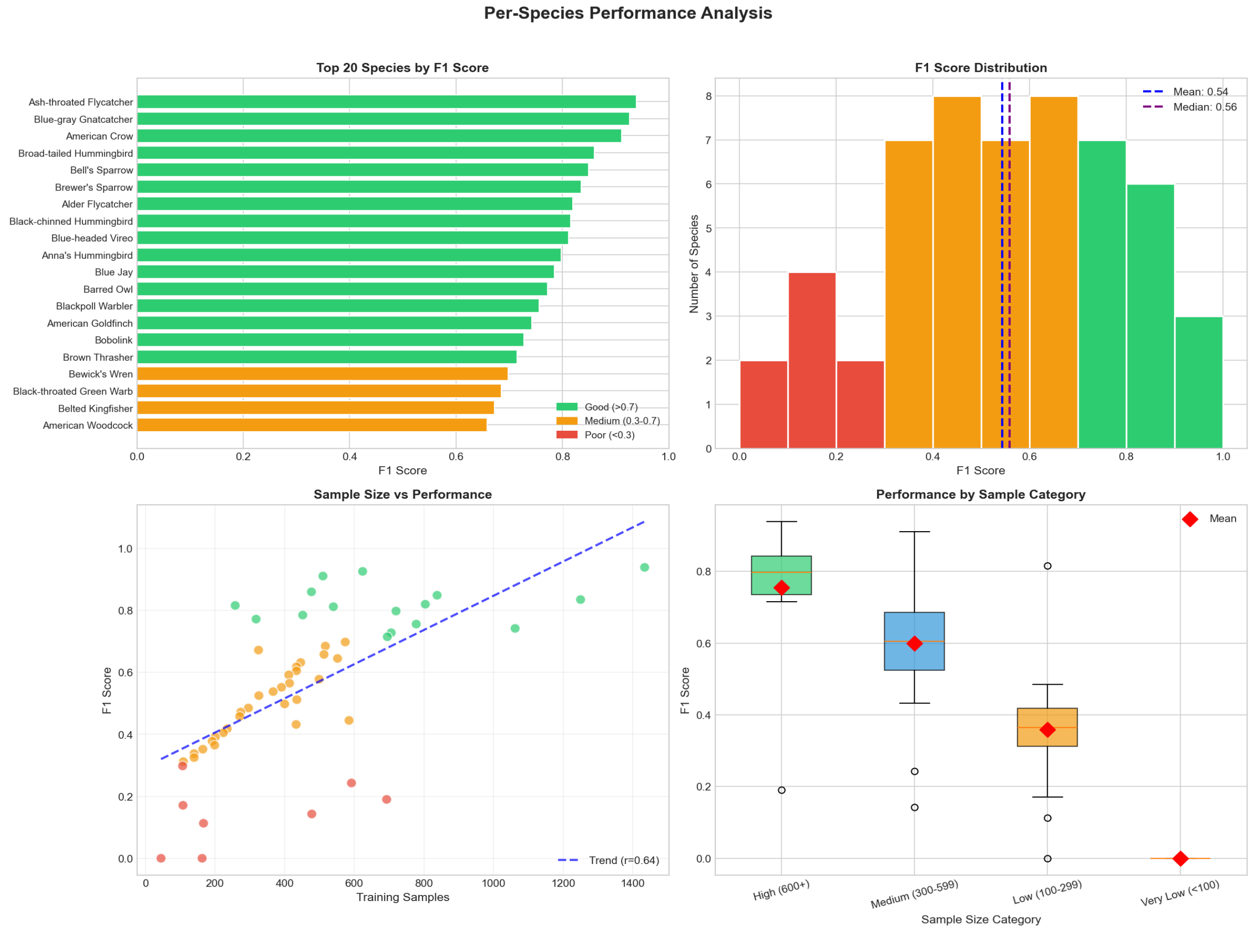

Figure 3: Species Performance Analysis

Category	Mean F1	Count
High (600+)	0.682	11
Medium (300-599)	0.624	24
Low (100-299)	0.503	17
Very Low (<100)	0.256	2

Table 5: Performance by Sample Category

6. Data Distribution

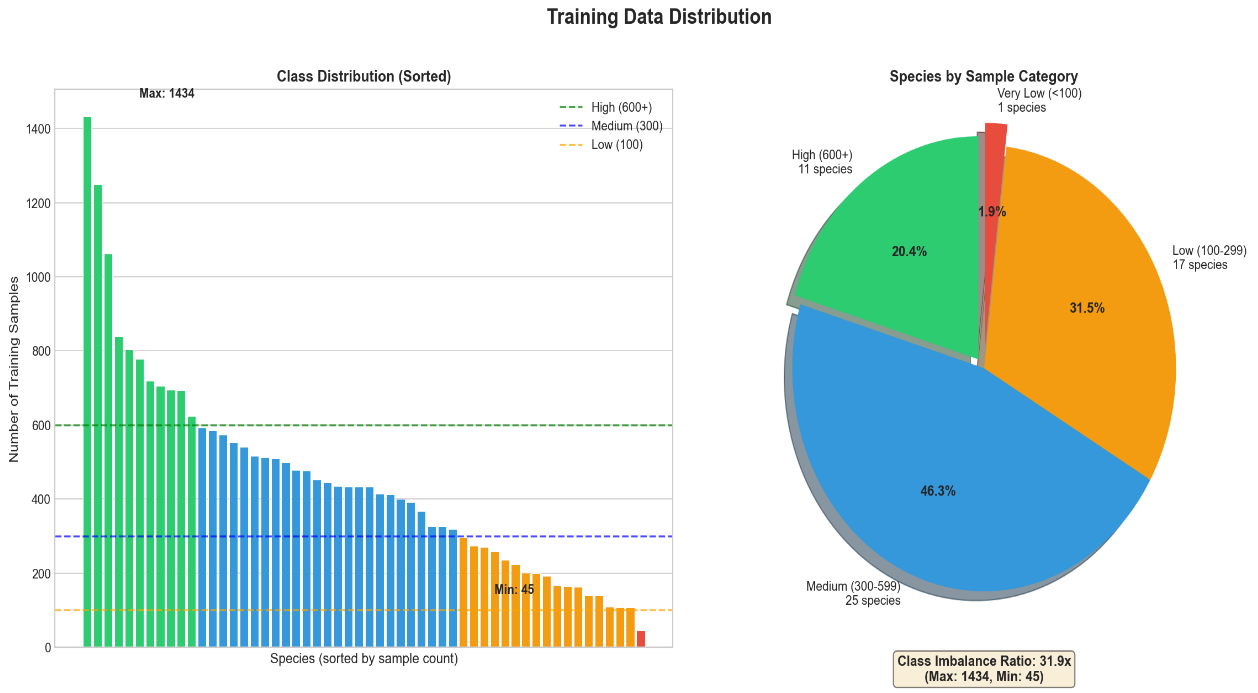

Figure 4: Training Data Distribution

Species	F1 Score
Ash-throated Flycatcher	0.939
Blue-gray Gnatcatcher	0.926
American Crow	0.911
Broad-tailed Hummingbird	0.860
Bell's Sparrow	0.849

Table 6: Top 5 Performing Species

Species	F1	Issue
Bufflehead	0.00	Only 45 samples
Barn Swallow	0.00	Confusion
Bald Eagle	0.11	Similar to Barred Owl
Blue Grosbeak	0.14	Acoustic similarity

Table 7: Challenging Species

7. Error Analysis

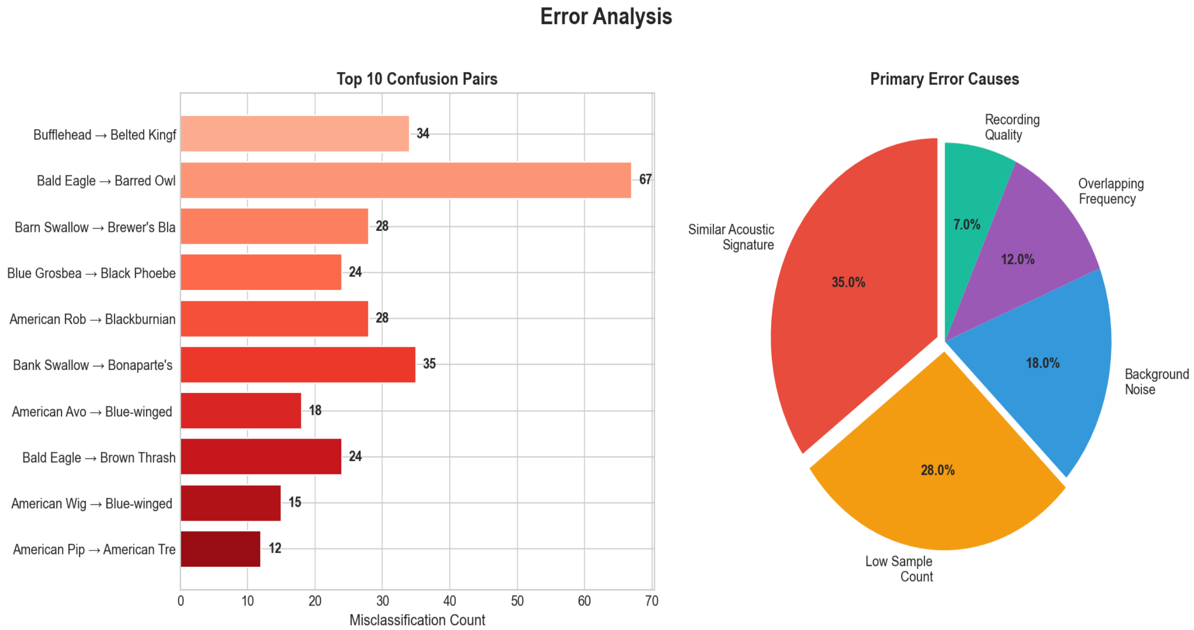

Figure 5: Error Analysis

Primary Error Causes:

- Similar Acoustic Signatures (35%): Overlapping frequency/call patterns
- Low Sample Count (28%): Insufficient training data for rare species
- Background Noise (18%): Environmental interference
- Overlapping Frequencies (12%): Species with similar vocal ranges

8. Conclusions

Key Achievements:
• Achieved 67.42% test accuracy on 54-class bird classification
• Top-5 accuracy of 82.57% demonstrates reliable species narrowing
• EfficientNet-B0 provides best efficiency (4.35M params vs 25.6M for ResNet-50)
• Fixed critical data leakage in original pipeline

Limitations:
• Two species (Bufflehead, Barn Swallow) achieve 0% F1
• Class imbalance affects rare species performance

Future Work:
• Collect more data for underrepresented species
• Implement multi-label classification
• Deploy as real-time monitoring system

References

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